SG No. 209 O^3 $F432$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$, $+\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `432`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `4b`, site symmetry: `432`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `8c`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 4: Wyckoff site: `24d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[0,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{4},\ 0,\ \frac{1}{4}\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4},\ 0,\ \frac{3}{4}\right]$	`[6,7,13,16]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[9,12,21,24]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[10,11,22,23]`

Table 5: Wyckoff site: `24e`, site symmetry: `4..`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,4,17,20]`
2	$[-x,\ 0,\ 0]$	`[2,3,18,19]`
3	$[0,\ x,\ 0]$	`[5,8,13,16]`
4	$[0,\ -x,\ 0]$	`[6,7,14,15]`
5	$[0,\ 0,\ x]$	`[9,12,22,23]`
6	$[0,\ 0,\ -x]$	`[10,11,21,24]`

Table 6: Wyckoff site: `32f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,9]`
2	$[-x,\ -x,\ x]$	`[2,7,12]`

continued ...

Table 6

No.	position	mapping
3	$[-x,\ x,\ -x]$	[3,8,10]
4	$[x,\ -x,\ -x]$	[4,6,11]
5	$[x,\ x,\ -x]$	[13,17,21]
6	$[-x,\ -x,\ -x]$	[14,19,24]
7	$[x,\ -x,\ x]$	[15,20,22]
8	$[-x,\ x,\ x]$	[16,18,23]

Table 7: Wyckoff site: 48g, site symmetry: ..2

No.	position	mapping
1	$[0,\ y,\ y]$	[1,18]
2	$[0,\ -y,\ y]$	[2,20]
3	$[0,\ y,\ -y]$	[3,17]
4	$[0,\ -y,\ -y]$	[4,19]
5	$[y,\ 0,\ y]$	[5,15]
6	$[y,\ 0,\ -y]$	[6,13]
7	$[-y,\ 0,\ y]$	[7,16]
8	$[-y,\ 0,\ -y]$	[8,14]
9	$[y,\ y,\ 0]$	[9,21]
10	$[-y,\ y,\ 0]$	[10,23]
11	$[y,\ -y,\ 0]$	[11,22]
12	$[-y,\ -y,\ 0]$	[12,24]

Table 8: Wyckoff site: 48h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{2},\ y,\ y\right]$	[1,18]
2	$\left[\frac{1}{2},\ -y,\ y\right]$	[2,20]
3	$\left[\frac{1}{2},\ y,\ -y\right]$	[3,17]
4	$\left[\frac{1}{2},\ -y,\ -y\right]$	[4,19]
5	$\left[y,\ \frac{1}{2},\ y\right]$	[5,15]
6	$\left[y,\ \frac{1}{2},\ -y\right]$	[6,13]
7	$\left[-y,\ \frac{1}{2},\ y\right]$	[7,16]
8	$\left[-y,\ \frac{1}{2},\ -y\right]$	[8,14]
9	$\left[y,\ y,\ \frac{1}{2}\right]$	[9,21]
10	$\left[-y,\ y,\ \frac{1}{2}\right]$	[10,23]
11	$\left[y,\ -y,\ \frac{1}{2}\right]$	[11,22]
12	$\left[-y,\ -y,\ \frac{1}{2}\right]$	[12,24]

Table 9: Wyckoff site: 48i, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[1,4]

continued ...

Table 9

No.	position	mapping
2	$\left[-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	[2,3]
3	$\left[\frac{1}{4},\ x,\ \frac{1}{4}\right]$	[5,8]
4	$\left[\frac{1}{4},\ -x,\ \frac{3}{4}\right]$	[6,7]
5	$\left[\frac{1}{4},\ \frac{1}{4},\ x\right]$	[9,12]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ -x\right]$	[10,11]
7	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	[13,16]
8	$\left[\frac{3}{4},\ -x,\ \frac{3}{4}\right]$	[14,15]
9	$\left[x,\ \frac{1}{4},\ \frac{3}{4}\right]$	[17,20]
10	$\left[-x,\ \frac{1}{4},\ \frac{1}{4}\right]$	[18,19]
11	$\left[\frac{1}{4},\ \frac{1}{4},\ -x\right]$	[21,24]
12	$\left[\frac{1}{4},\ \frac{3}{4},\ x\right]$	[22,23]

Table 10: Wyckoff site: 96j, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[x,\ -y,\ -z]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[z,\ -x,\ -y]$	[6]
7	$[-z,\ -x,\ y]$	[7]
8	$[-z,\ x,\ -y]$	[8]
9	$[y,\ z,\ x]$	[9]
10	$[-y,\ z,\ -x]$	[10]
11	$[y,\ -z,\ -x]$	[11]
12	$[-y,\ -z,\ x]$	[12]
13	$[y,\ x,\ -z]$	[13]
14	$[-y,\ -x,\ -z]$	[14]
15	$[y,\ -x,\ z]$	[15]
16	$[-y,\ x,\ z]$	[16]
17	$[x,\ z,\ -y]$	[17]
18	$[-x,\ z,\ y]$	[18]
19	$[-x,\ -z,\ -y]$	[19]
20	$[x,\ -z,\ y]$	[20]
21	$[z,\ y,\ -x]$	[21]
22	$[z,\ -y,\ x]$	[22]
23	$[-z,\ y,\ x]$	[23]
24	$[-z,\ -y,\ -x]$	[24]